

Learn more about the Caesar cipher at https://en.wikipedia.org/wiki/Caesar_cipher. If you'd like to learn about ciphers and code breaking in general, you can read my book *Cracking Codes with Python* (No Starch Press, 2018; <https://nostarch.com/crackingcodes/>).

The Program in Action

When you run `caesarcipher.py`, the output will look like this:

```
Caesar Cipher, by Al Sweigart al@inventwithpython.com
Do you want to (e)ncrypt or (d)ecrypt?
> e
Please enter the key (0 to 25) to use.
> 4
Enter the message to encrypt.
> Meet me by the rose bushes tonight.
QIIX QI FC XLI VSWI FYWLIW XSRMKLX.
Full encrypted text copied to clipboard.
```

```
Caesar Cipher, by Al Sweigart al@inventwithpython.com
Do you want to (e)ncrypt or (d)ecrypt?
> d
Please enter the key (0 to 26) to use.
> 4
Enter the message to decrypt.
> QIIX QI FC XLI VSWI FYWLIW XSRMKLX.
MEET ME BY THE ROSE BUSHES TONIGHT.
Full decrypted text copied to clipboard.
```

How It Works

Like most cipher programs, the Caesar cipher works by translating characters into numbers, performing some math operations on those numbers, and translating the numbers back into text characters. In the context of ciphers, we call these text characters *symbols*. Symbols can include letters, numeric digits, and punctuation marks, each of which gets assigned a unique integer. In the case of the Caesar cipher program, the symbols are all letters, and their integers are their position in the `SYMBOLS` string: `'ABCDEFGHIJKLMNOPQRSTUVWXYZ'`.

```
1. """Caesar Cipher, by Al Sweigart al@inventwithpython.com
2. The Caesar cipher is a shift cipher that uses addition and subtraction
3. to encrypt and decrypt letters.
4. More info at: https://en.wikipedia.org/wiki/Caesar_cipher
5. View this code at https://nostarch.com/big-book-small-python-projects
6. Tags: short, beginner, cryptography, math"""
7.
8. try:
9.     import pyperclip # pyperclip copies text to the clipboard.
10. except ImportError:
11.     pass # If pyperclip is not installed, do nothing. It's no big deal.
```